

ICPSR/Retraction Watch, First Look

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DOIs available?

```
# What proportion of entries in each table have DOIs (ICPSR, Retraction Watch)
c(length(icpsr$DOI[which(!is.na(icpsr$DOI))]) / length(icpsr$DOI),
  length(retractions$OriginalPaperDOI[which(!is.na(retractions$OriginalPaperDOI))]) /
  length(retractions$OriginalPaperDOI))
```

```
## [1] 0.4917015 0.9444636
```

```
# Duplicate DOIs in retractions?
```

```
retractions |>
  drop_na(OriginalPaperDOI) |>
  add_count(OriginalPaperDOI) |>
  filter(n > 1) |>
  distinct() |>
  select("Record.ID", "Title", "OriginalPaperDOI", "n") |>
  nrow()
```

```
## [1] 10821
```

```
# Retraction Watch database contains 'duplicate' DOIs - if a paper
# was corrected then retracted, there are separate entries for the
# correction and the retraction
```

```
# pivot_wider before merging?
```

```
# How many entries for each type of retraction reason?
```

```
retractions |>
  drop_na(RetractionNature) |>
  group_by(RetractionNature) |>
  summarize(n())
```

```
## # A tibble: 4 x 2
##   RetractionNature   `n()`
##   <chr>             <int>
## 1 Correction         1362
## 2 Expression of concern 3440
## 3 Reinstatement      158
## 4 Retraction        61681
```

```
# Duplicate DOIs in ICPSR?
```

```
icpsr |>
  drop_na(DOI) |>
```

```
add_count(DOI) |>  
filter(n > 1) |>  
distinct() |>  
select("Reference.ID", "Title", "DOI", "n") |>  
nrow()
```

```
## [1] 0
```

```
# Doesn't look like it! Thank you data librarians :)
```

RIS Types

```
# Article type found in 'RIS.Type' column, not 'Article.Type'

# For entries without DOIs in ICPSR, what type of materials are they?

# 'RIS.Type' available for what proportion of icpsr entries?
icpsr |>
  drop_na(RIS.Type) |>
  select("Reference.ID", "Title", "DOI", "RIS.Type") |>
  nrow() / nrow(icpsr)
```

```
## [1] 1

# 'RIS.Type' available for what proportion of icpsr entries
# that have a DOI?
icpsr |>
  drop_na(DOI) |>
  drop_na(RIS.Type) |>
  select("Reference.ID", "DOI", "Type.of.Work") |>
  nrow() / nrow(filter(icpsr, is.na(DOI)))
```

```
## [1] 0.9673478

# ... that don't have a DOI?
icpsr |>
  filter(is.na(DOI)) |>
  drop_na(RIS.Type) |>
  select("Reference.ID", "Title", "DOI", "Type.of.Work") |>
  nrow() / nrow(filter(icpsr, is.na(DOI)))
```

```
## [1] 1

# 'Type.of.Work' appears to be available for < 10% of entries -
# around 5.4% of entries with a DOI and 18.5% of those w/o a DOI

# TODO: fix NA values - diff NA values associated w diff columns

# How many entries associated with each type of work?
icpsr |>
  drop_na(RIS.Type) |>
  filter(RIS.Type != "(unknown)") |>
  group_by(RIS.Type) |>
  summarize(count = n()) |>
  arrange(desc(count))
```

```
## # A tibble: 12 x 2
##   RIS.Type count
##   <chr>    <int>
## 1 JOUR      70409
## 2 RPRT      15470
## 3 CONF      11533
## 4 THES      10349
## 5 CHAP       5148
## 6 BOOK       2809
## 7 GEN        1387
```

```
## 8 MANSCPT      686
## 9 NEWS          575
## 10 ELEC         548
## 11 MGZN         340
## 12 ADVS         225
```

Needs to be standardized .. capitalization differences interfering

'ArticleType' unavailable for what proportion of RW entries?

```
retractions |>
  select(Record.ID, ArticleType) |>
  drop_na(ArticleType) |>
  nrow()
```

```
## [1] 66641
```

ArticleType available for all RW entries, it looks like!

How many entries associated with each type of work in RW?

```
retractions |>
  select(Record.ID, ArticleType) |>
  drop_na(ArticleType) |>
  unnest(c(ArticleType)) |>
  group_by(ArticleType) |>
  summarize(count = n()) |>
  arrange(desc(count))
```

```
## # A tibble: 26 x 2
##   ArticleType      count
##   <chr>          <int>
## 1 Research Article 45084
## 2 Conference Abstract/Paper 13280
## 3 Clinical Study  2970
## 4 Review Article  2581
## 5 Meta-Analysis   875
## 6 Case Report     868
## 7 Book Chapter/Reference Work 576
## 8 Article in Press 519
## 9 Letter          517
## 10 Commentary/Editorial 443
## # i 16 more rows
```

These overlap - ArticleType is a list-type column

Publishers

```
# Publication name available for what proportion of entries?
```

```
# Secondary.Title includes publication name if entry is  
# a journal/newspaper
```

```
icpsr |>  
  select(Reference.ID, Secondary.Title) |>  
  drop_na(Secondary.Title) |>  
  nrow() / nrow(icpsr)
```

```
## [1] 0.8259025
```

```
# Now looking at Publisher - contains info on gov't  
# bureaus etc responsible for the publication  
# Approx equivalent to 'Institution' in RW data?
```

```
icpsr |>  
  select(Reference.ID, Publisher) |>  
  drop_na(Publisher) |>  
  nrow() / nrow(icpsr)
```

```
## [1] 0.3594607
```

```
# Around 36% of icpsr entries list the publisher
```

```
# What about those entries without DOI?
```

```
# Secondary.Title
```

```
icpsr |>  
  filter(is.na(DOI)) |>  
  select(Reference.ID, Secondary.Title) |>  
  drop_na(Secondary.Title) |>  
  nrow() / nrow(filter(icpsr, is.na(DOI)))
```

```
## [1] 0.6794059
```

```
# Publisher
```

```
icpsr |>  
  filter(is.na(DOI)) |>  
  select(Reference.ID, Publisher) |>  
  drop_na(Publisher) |>  
  nrow() / nrow(filter(icpsr, is.na(DOI)))
```

```
## [1] 0.6396239
```

```
# Around 64% of entries without a DOI have a publisher listed
```

```
# What publications appear frequently in ISPCR?
```

```
# Secondary.Title
```

```
icpsr |>  
  select(Reference.ID, Secondary.Title) |>  
  drop_na(Secondary.Title) |>  
  group_by(Secondary.Title) |>  
  summarize(count = n()) |>  
  arrange(desc(count))
```

```
## # A tibble: 19,705 x 2
##   Secondary.Title                                count
##   <chr>                                           <int>
## 1 ProQuest Dissertations and Theses              1021
## 2 Journal of Marriage and Family                  971
## 3 American Journal of Public Health               842
## 4 Journals of Gerontology, Series B: Psychological Sciences and Social S~ 649
## 5 Social Forces                                  635
## 6 Social Science Quarterly                       627
## 7 annual meeting of the American Political Science Association 618
## 8 American Sociological Review                   586
## 9 American Journal of Political Science           583
## 10 American Political Science Review              556
## # i 19,695 more rows
```

```
# Publisher
```

```
icpsr |>
  select(Reference.ID, Publisher) |>
  drop_na(Publisher) |>
  group_by(Publisher) |>
  summarize(count = n()) |>
  arrange(desc(count))
```

```
## # A tibble: 6,988 x 2
##   Publisher                                count
##   <chr>                                   <int>
## 1 American Society of Criminology           965
## 2 United States Department of Justice, National Institute of Justice 879
## 3 U.S. Department of Health and Human Services, Administration for Child~ 631
## 4 National Bureau of Economic Research       603
## 5 United States Department of Justice, Bureau of Justice Statistics 600
## 6 American Sociological Association           573
## 7 United States Department of Education, Office of Educational Research ~ 448
## 8 University of Michigan                    428
## 9 Substance Abuse and Mental Health Services Administration 420
## 10 University of North Carolina at Chapel Hill, Carolina Population Center 419
## # i 6,978 more rows
```

```
# Publication name available for what proportion of entries in RW?
```

```
# Secondary.Title includes publication name if entry is
# a journal/newspaper
```

```
retractions |>
  select(Record.ID, Journal) |>
  drop_na(Journal) |>
  nrow() / nrow(retractions)
```

```
## [1] 1
```

```
# All entries list a 'Journal' - column just holds the source
# of the article (incl. journals, books, serials, etc)
```

```
# What journals are most common in RW?
```

```
retractions |>
  select(Record.ID, Journal) |>
```

```
drop_na(Journal) |>
unnest(c(Journal)) |>
group_by(Journal) |>
summarize(count = n()) |>
arrange(desc(count))
```

```
## # A tibble: 8,496 x 2
##   Journal                                count
##   <chr>                                <int>
## 1 Journal of Intelligent & Fuzzy Systems      1566
## 2 2011 International Conference on E-Business and E-Government (ICEE)    1280
## 3 PLoS One                                  1224
## 4 2011 5th International Conference on Bioinformatics and Biomedical Eng~ 1084
## 5 Journal of Healthcare Engineering          1074
## 6 Computational and Mathematical Methods in Medicine    1067
## 7 Computational Intelligence and Neuroscience    1028
## 8 BioMed Research International              953
## 9 Security and Communication Networks          949
## 10 Journal of Physics: Conference Series        878
## # i 8,486 more rows
```

Merging

```
# Initial brief attempt at a merge
```

```
merged <- inner_join(
  x = (icpsr %>%
    filter(!is.na(DOI)) %>%
    select(Reference.ID, Title, DOI) %>%
    unnest(c(DOI))),
  y = (retractions %>%
    filter(!is.na(OriginalPaperDOI)) %>%
    select(Record.ID, Title, OriginalPaperDOI, RetractionDate, RetractionNature, Reason) %>%
    unnest(c(OriginalPaperDOI))),
  by = join_by(DOI == OriginalPaperDOI)
)

print(merged)
```

```
## # A tibble: 9 x 8
##   Reference.ID Title.x   DOI   Record.ID Title.y RetractionDate RetractionNature
##   <int> <chr>    <chr> <list>    <list>    <list>         <chr>
## 1    11313 Worlds A~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 2    15930 Nonmetro~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 3    29220 School s~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 4     87821 Suicide ~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 5    140614 Electron~ 10.1~ <chr [1]> <chr>    <chr [1]>    Correction
## 6    140614 Electron~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 7    142038 The verd~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 8    144102 Chronic ~ 10.1~ <chr [1]> <chr>    <chr [1]>    Correction
## 9    155116 Tuning i~ 10.1~ <chr [1]> <chr>    <chr [1]>    Expression of c~
## # i 1 more variable: Reason <list>
```

```
# Is 9 observations around what we'd expect??  
# Slightly concerning might be a data cleaning problem  
  
# remove duplicate row from merged  
merged = merged[c(1:4, 6:9),]
```


Citation Data

Change date strings to lubridate Date type

```
make_date = function(str) {
  # month, day, year format
  # day may or may not be included
  date_strs = unlist(strsplit(str, " "))

  # account for date ranges
  # TEMP - if a range is given, take the earliest given date
  if (str_length(date_strs[1]) > 3) {
    date_strs[1] <- strsplit(date_strs[1], "-")[1][1]
  }
  if (str_length(date_strs[length(date_strs)]) > 4) {
    date_strs[length(date_strs)] <-
      strsplit(date_strs[length(date_strs)], "-")[1][1]
  }
  if (length(date_strs) > 2 & str_length(date_strs[2]) > 2) {
    date_strs[2] <- strsplit(date_strs[2], "-")[1][1]
  }

  # QUESTION: What to do if day of publication is not provided?
  date_nums = c(
    # year
    as.character(date_strs[length(date_strs)]),
    # month
    as.character(which(date_strs[1] == str_to_upper(month.abb))),
    # day
    # TEMP FIX - use the first of the month if no day provided
    ifelse(length(date_strs) > 2, as.character(date_strs[2]), "1")
  )

  return(ymd(paste0(date_nums[1], "-", date_nums[2], "-", date_nums[3])))
}

# read data - drop leading 10 rows
# drop citations by year data - will reconstru
process.tbl <- function(filename) {
  df <- read.csv(filename, skip = 10)
  df <- subset(df, select = c(1:21))
  df |> group_by(Publication.Year) |>
    summarise(Num.Citations = n()) -> cit_per_year
  return(cit_per_year)
}

# order corresponds to order of rows in merged
tblnames = c("citation-reports/WorldsApart.csv",
             "citation-reports/NonmetroRole.csv",
             "citation-reports/SchoolSocialBonds.csv",
             "citation-reports/SuicideNatDisasters.csv",
             "citation-reports/ECigUse.csv",
             "citation-reports/HowDidTheyDecide.csv",
             "citation-reports/MarijuanaUse.csv",
             "citation-reports/TurningIn.csv")
```

```

    )

# set console width
options(width = 80)

get.proportions = function(filename) {
  # load in table
  tbl = process.tbl(filename)

  # get retraction year
  ix = which(tblnames == filename)
  mdy_hm(merged$RetractionDate[[ix]]) |>
    year() -> ret_year

  # number of citations pre- and post- retraction
  # excludes citations from the same year - more work needed to try to use
  # day/month for comparison
  pre_ret = (sum(tbl$Num.Citations[which(tbl$Publication.Year < ret_year)]) /
    sum(tbl$Num.Citations))
  post_ret = (sum(tbl$Num.Citations[which(tbl$Publication.Year > ret_year)]) /
    sum(tbl$Num.Citations))

  return(c(pre_ret, post_ret))
}

# plot citations over time
plot.citations = function(filename) {
  tbl = process.tbl(filename)
  mdy_hm(merged$RetractionDate[[ix]]) |>
    year() -> ret_year

  # get xlim for plot - make sure retraction
  # year is included
  min_x = min(
    ret_year,
    min(tbl$Publication.Year)
  )
  max_x = max(
    ret_year,
    max(tbl$Publication.Year)
  )

  plot(x = process.tbl(tblnames[ix])$Publication.Year,
    y = process.tbl(tblnames[ix])$Num.Citations,
    type = 'o', pch = 20,
    main = strwrap(paste0("Citations of ", "\"",
      merged$Title.x[ix], "\""),
      width = 50),
    xlab = "Publication Year",
    ylab = "Number of Citations",
    xlim = c(min_x, max_x)
  )
  abline(v = mdy_hm(merged$RetractionDate[[ix]]) %>% year(),
    lwd = 2, col = "red")
}

```

```

cplot = recordPlot()

return(cplot)
}

# setwd(...)

for (ix in 1:length(merged)) {
  # print proportion of citations from before,
  # after retraction year
  prop_citations = get.proportions(tblnames[ix])
  print(strwrap(merged$Title.x[ix], width = getOption("width")))
  cat("Pre-retraction:", prop_citations[1],
      ", Post-retraction", prop_citations[2], "\n",
      fill = FALSE)

  # plot citations
  print(plot.citations(tblnames[ix]))
}

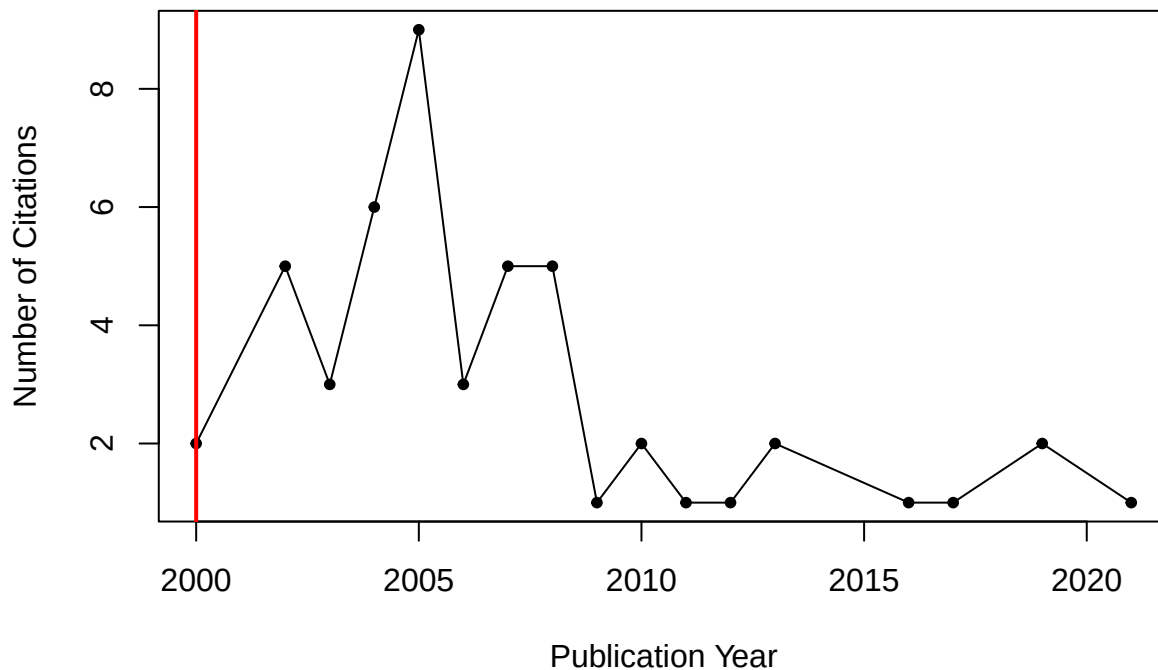
```

```

## [1] "Worlds Apart? The Reception of Genetically Modified Foods in Europe and the"
## [2] "U.S."
## Pre-retraction: 0 , Post-retraction 0.96

```

Citations of "Worlds Apart? The Reception of Genetically Modified Foods in Europe and the U.S."

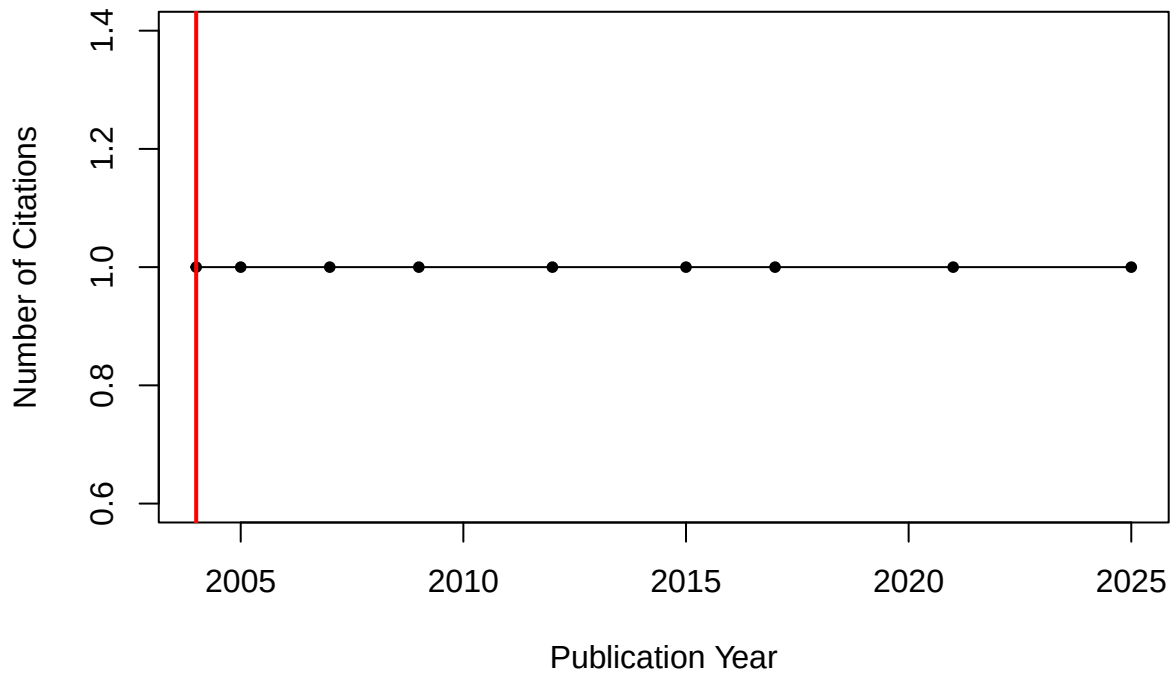


```

## [1] "Nonmetropolitan sex-role ideologies: A longitudinal study"
## Pre-retraction: 0 , Post-retraction 0.8888889

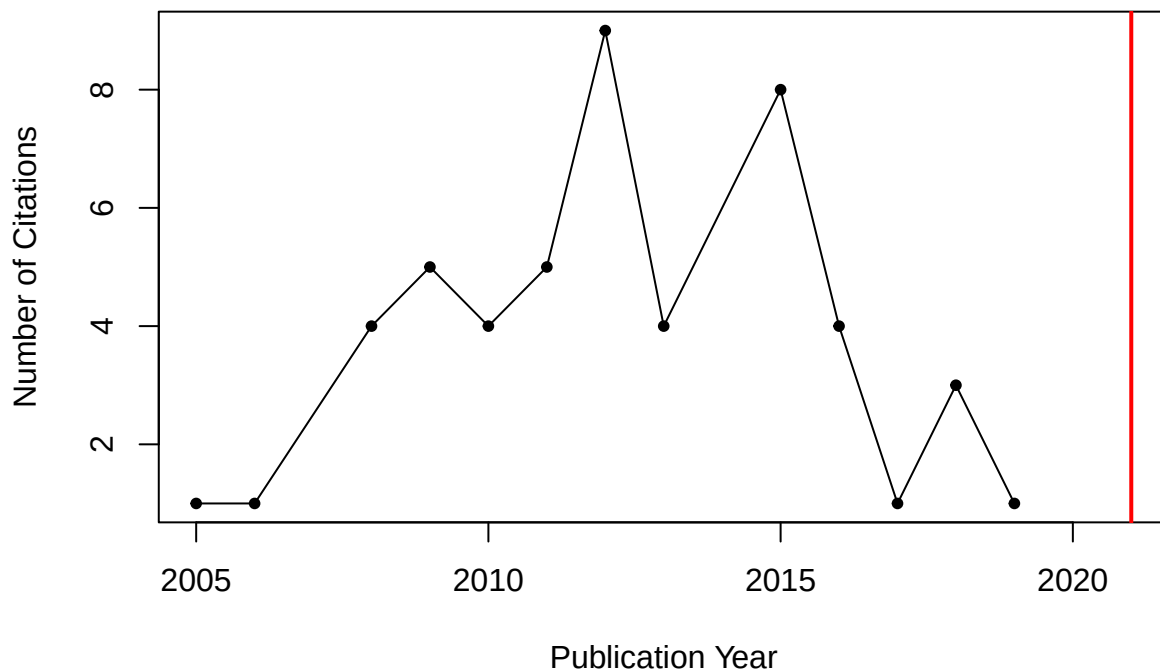
```

Citations of "Nonmetropolitan sex-role ideologies: A longitudinal study"



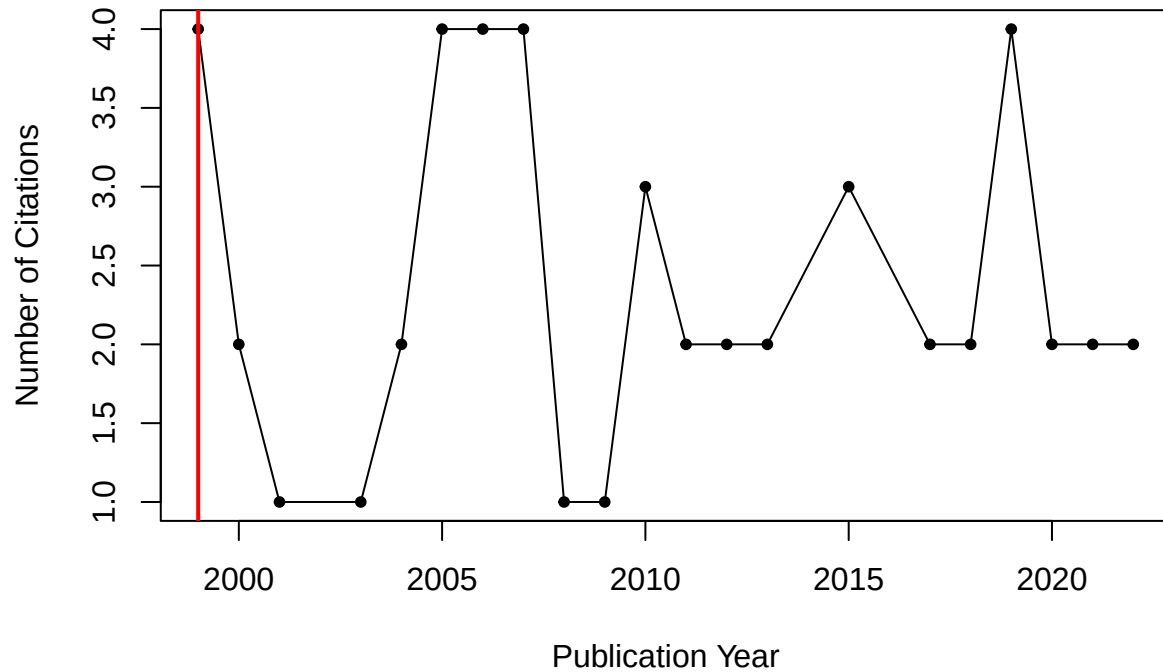
```
## [1] "School social bonds, school climate, and school misbehavior: A multilevel"  
## [2] "analysis"  
## Pre-retraction: 1 , Post-retraction 0
```

Citations of "School social bonds, school climate, and school misbehavior: A multilevel analysis"



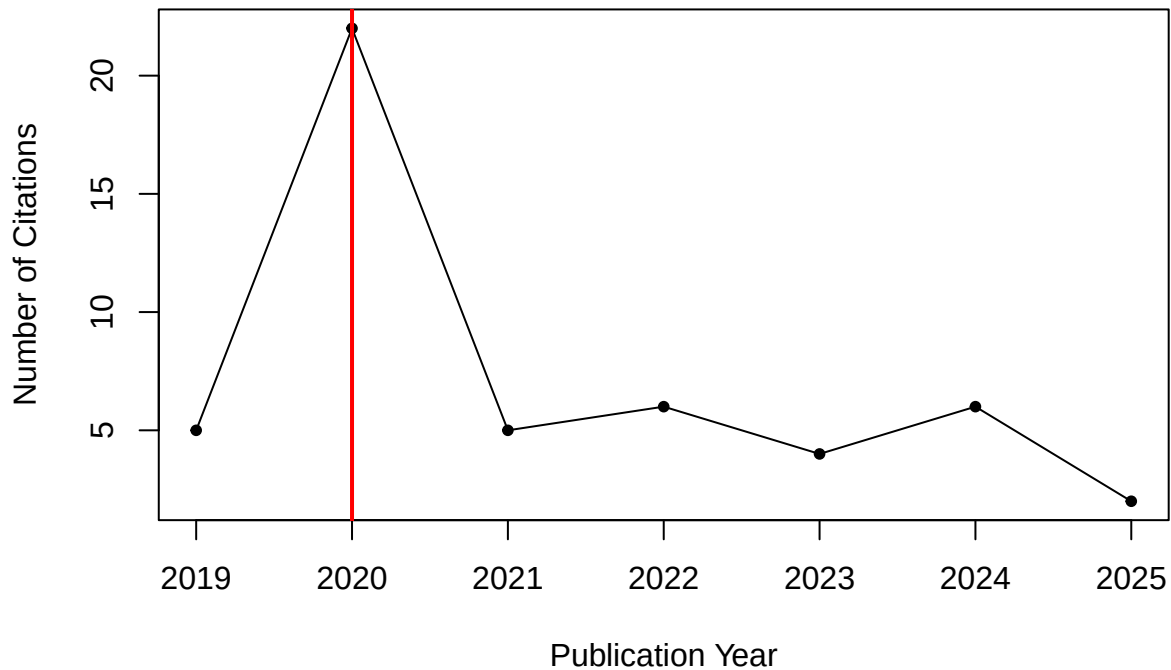
```
## [1] "Suicide after natural disasters"  
## Pre-retraction: 0 , Post-retraction 0.92
```

Citations of "Suicide after natural disasters"



```
## [1] "Electronic cigarette use and myocardial infarction among adults in the US"  
## [2] "Population Assessment of Tobacco and Health"  
## Pre-retraction: 0.1 , Post-retraction 0.46
```

Citations of "Electronic cigarette use and myocardial infarction among adults in the US Population Assessment of Tobacco and Health"

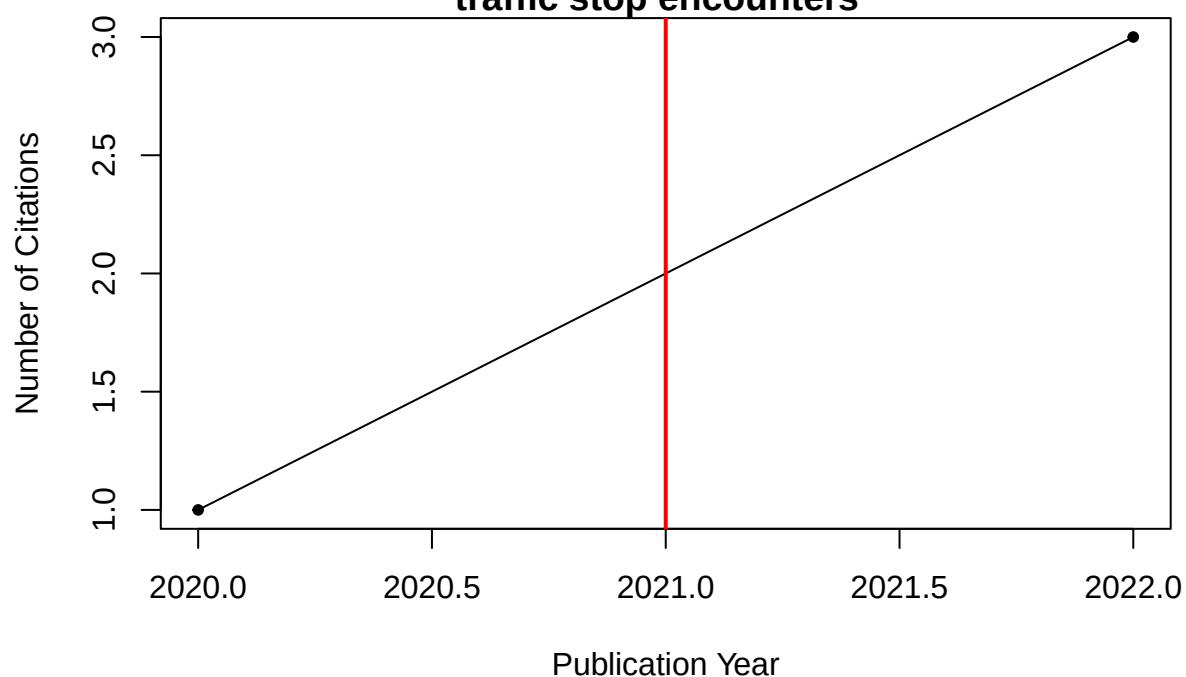


```
## Warning in read.table(file = file, header = header, sep = sep, quote = quote, :
## incomplete final line found by readTableHeader on
## 'citation-reports/HowDidTheyDecide.csv'

## [1] "The verdict is in: How did they decide? Using drivers' self-reported data to"
## [2] "understand officers' decision making during traffic stop encounters"
## Pre-retraction: 0.25 , Post-retraction 0.75

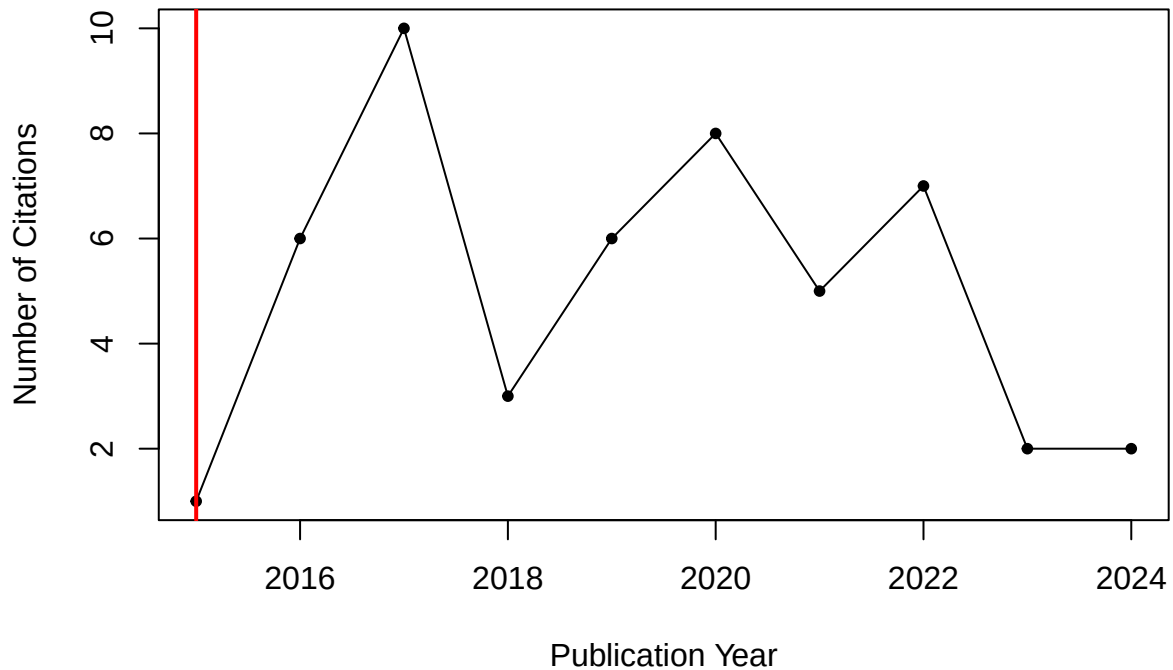
## Warning in read.table(file = file, header = header, sep = sep, quote = quote, :
## incomplete final line found by readTableHeader on
## 'citation-reports/HowDidTheyDecide.csv'
## Warning in read.table(file = file, header = header, sep = sep, quote = quote, :
## incomplete final line found by readTableHeader on
## 'citation-reports/HowDidTheyDecide.csv'
## Warning in read.table(file = file, header = header, sep = sep, quote = quote, :
## incomplete final line found by readTableHeader on
## 'citation-reports/HowDidTheyDecide.csv'
```

Citations of "The verdict is in: How did they decide? Using drivers' self-reported data to understand officers' decision making during traffic stop encounters"



```
## [1] "Chronic adolescent marijuana use as a risk factor for physical and mental"
## [2] "health problems in young adult men"
## Pre-retraction: 0 , Post-retraction 0.98
```

Citations of "Chronic adolescent marijuana use as a risk factor for physical and mental health problems in young adult men"



```
## [1] "Tuning in, not turning out: Evaluating the impact of ethnic television on"
## [2] "political participation"
## Pre-retraction: 0 , Post-retraction 0.9285714
```

Citations of "Tuning in, not turning out: Evaluating the impact of ethnic television on political participation"

